Akshat Rauthan

Haridwar | akshatrauthan9433@gmail.com | +91 72175 39649 | akshat-rauthan | AkshatRauthan

Career Objective

Backend engineer who owned the production backend of a trading analytics platform serving 1k+ users. Experienced in microservices, real-time systems (WebSockets, Kafka), and distributed infrastructure.

Education

COER University – *B.Tech in Computer Science & Engineering* | **CGPA:** 8.93/10 *July 2023 – Jun 2027*

Internship

Tradylytics - Software Developer Intern [Company Link] *July 2025 – Aug 2026*

- Primary owner of the **backend infrastructure** for a trading analytics platform serving **1k+ users**, working in a four-person engineering team: designed services, databases, and third-party integrations end to end.
- Led the migration of the legacy **JavaScript codebase to TypeScript** and re-architected the backend into **two services**, eliminating logic duplicated across multiple places.
- Redesigned database schema to pre-calculate rolling trade metrics, reducing **API response time by 30%** .
- Integrated **10+ third-party brokerage platforms** leveraging **Webhooks and WebSockets** to build a centralized real-time portfolio dashboard covering **equity, futures, options and crypto** investements.
- Built a **multi-tenant B2B platform** for trading schools, enabling institutional partners to provision student access and monitor cohort portfolios via a centralized admin portal providing access to our AI integrated journaling platform.
- **Tech Stack:** Node.js, JavaScript, TypeScript, FastAPI, Python, MongoDB, Redis, Docker

Projects

AeroLink - Flight Booking Application [GitHub] *July 2026*

- Architected a **7-service microservices backend** (Node.js, TypeScript, **pnpm monorepo**, Docker) fronted by a custom **reverse proxy** handling auth, **IP + path-based rate limiting**, and inline **gRPC** token validation.
- Engineered a **real-time flight data streaming pipeline** in Go, generating synthetic route, inventory, and dynamic pricing streams over **Apache Kafka** via CRON-driven routines to mimic live airline partner APIs.
- Designed a **plug-and-play database sharding and read/write replication layer** in the shared infrastructure module, enabling any service to independently opt into horizontal scaling and read distribution without modifying business logic.
- Implemented **RabbitMQ** workflows using **TTL queues and dead-letter exchanges** for automatic seat-hold expiry and notification delivery with **exponential backoff retries and a Dead Letter Queue** for failed messages.
- **Tech Stack:** Node.js, TypeScript, Go, MySQL, Redis, RabbitMQ, Kafka, gRPC, Docker

Vault - Document Integrity System [GitHub] *October 2026 - Present*

- Designed a **tamper-detection system for PDFs** using **SHA-256 hashing, Merkle trees, and Ed25519 digital signatures**, pinpointing the exact **page and paragraph** that was modified instead of returning a generic "tampered" result.
- Built a **deterministic, versioned segmentation and canonicalization pipeline** (pdf.js) so identical content always yields identical hashes, using domain-separated Merkle nodes to prevent second-preimage attacks and detect reordering.
- Implemented **signed, versioned proofs** with a portable sidecar file and backend recovery copy, plus a React viewer that highlights modified paragraphs on the original PDF.
- Integrated **OpenBao** for key management: user private keys are stored in it and manifests are signed inside it, so keys never reach the application layer. Secured the backend with **TLS and internally issued certificates** for its setup.
- **Tech Stack:** TypeScript, Node.js, Express, React, MongoDB, pdf.js, OpenBao

Real-Time Chat App [GitHub] *April 2026*

- Built a horizontally scalable **WebSocket backend** (Socket.IO, **Redis Pub/Sub**) with a TTL-based Redis presence and session store, relaying messages across multiple nodes behind Nginx.
- Decoupled delivery from persistence with a **write-behind batch queue**, cutting **P99 message latency from 85ms to less than 20ms**; replaced N+1 queries with batch queries, CTEs, and composite indexes, reducing **DB IOPS by 75%**.
- Load-tested with **k6 (5,000 concurrent WebSockets)** and exposed **Prometheus** metrics; added **E2EE** (TweetNaCl) and **WebRTC** signaling for voice/video calls.
- **Tech Stack:** TypeScript, Node.js, Socket.IO, Redis, BullMQ, PostgreSQL, Docker, React

Technical Skills

Languages: C++, TypeScript, Go, Python

Databases: MongoDB, MySQL, PostgreSQL

Backend: WebSockets, gRPC, Kafka, RabbitMQ

DevOps: Docker, Kubernetes, Github Actions